

# 1 diffblas: algorithmically differentiated BLAS routines

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DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

## Software

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Submitted: 01 January 1970

Published: unpublished

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## 6 Summary

7 diffblas is a library that provides BLAS routines algorithmically differentiated using algorithmic  
8 differentiation principles ([Griewank & Walther, 2008](#)) from their reference implementations in  
9 [LAPACK](#) on GitHub using the automatic differentiation tool Tapenade ([Hascoët & Pascual,](#)  
10 [2013](#)). It supports four modes: tangent (`_d`), vector tangent (`_dv`), adjoint (`_b`), and vector  
11 adjoint (`_bv`).

12 In addition to differentiating the standard Fortran-style BLAS interface, diffblas also provides  
13 differentiated CBLAS routines, facilitating interoperability with C and other languages. Its API  
14 mirrors BLAS / CBLAS, with additional arguments specifying differentiation variables, making  
15 integration into existing workflows straightforward.

16 diffblas relies on the underlying standard BLAS implementation and is agnostic to the  
17 backend (OpenBLAS, BLIS, MKL, Apple Accelerate, or libblastrampoline), ensuring both  
18 high performance and portability. Precompiled artifacts are available on GitHub, simplifying  
19 installation and use across different programming environments.

20 By providing efficient and accurate derivatives of linear algebra operations, diffblas facilitates  
21 gradient-based optimization, sensitivity analysis, and derivative-based workflows in scientific  
22 computing.

## 23 Statement of need

24 Linear algebra routines such as those in LAPACK are widely used in scientific computing,  
25 optimization, and machine learning. However, they do not provide derivatives, which are often  
26 required for gradient-based algorithms.

27 diffblas addresses this gap by providing algorithmically differentiated BLAS routines directly  
28 from reference LAPACK implementations and following relevant differentiation rules ([Giles,](#)  
29 [2008](#)).

## 30 State of the field

31 Automatic source-to-source differentiation tools, such as Tapenade ([Hascoët & Pascual,](#)  
32 [2013](#)), ADOL-C ([Griewank et al., 1996](#)), or TAF ([Giering & Kaminski, 2003](#)), provide general  
33 mechanisms to compute derivatives of code.

34 ([Jonasson et al., 2020](#)) derives scalar reverse mode derivative formulae for BLAS routines and  
35 provides their Fortran code. Such an approach is generally more efficient than differentiating  
36 through the BLAS routines using an AD tool ([Jonasson et al., 2020](#)). They do not however  
37 consider the forward mode or vector modes.

Derivatives of linear algebra routines are available in Python packages such as JAX (?) and PyTorch (?). These implementations are not available outside these frameworks. Enzyme and Enzyme.jl perform optimized BLAS and CuBLAS differentiation at the MLIR level(?) but the derivative code is not available externally. ...

## Research impact statement

This work was inspired in part by a need to differentiate a Fortran code (Hascoët et al., 2026) that uses BLAS and LAPACK routines, and to use the differentiated application for gradient-based optimization.

Providing both the standard and CBLAS interfaces ensures that diffblas can be adopted across different programming environments, facilitating derivative computations in diverse scientific computing projects. Precompiled artifacts on GitHub further simplify integration, enabling rapid deployment in multiple languages and scientific computing projects.

## Acknowledgements

This work was supported in part by the Applied Mathematics activity within the U.S. Department of Energy, Office of Science, Office of Advanced Scientific Computing Research Applied Mathematics, and Office of Nuclear Physics SciDAC program under Contract No. DE-AC02-06CH11357. This work was supported in part by NSF CSSI grant 2104068.

## AI usage disclosure

Generative AI was used to ... ChatGPT was used to check spelling, grammar, and clarity of the English text in this paper.

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